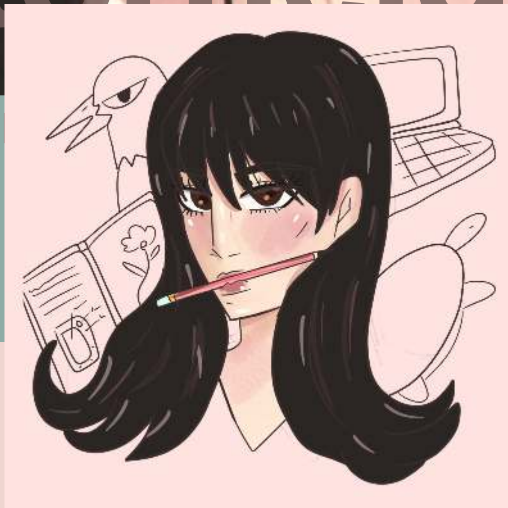

(PROBABLY ALMOST)

EVERYTHING ON THE SYLLABUS

XBI11 IAL EDEXCEL BIOLOGY

UNIT 1 & 2

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Cell Division

- What is cell division? How does this differ for prokaryotes/eukaryotes?

  Cell division is the process by which new cells are formed and genetic information is inherited. For prokaryotes or simple cells, binary fission tends to occur while for complex cells, they undergo mitosis or meiosis.

- Describe a chromosome, chromatin, sister chromatids & telomeres.

  Chromosome

  A chromosome is a rod shaped structure that contains DNA, or RNA and protein. The DNA contains genes (short segments) that code for particular proteins and characteristics in the body. The location of a gene on a chromosome is known as the **locus**. All the genes together make your body's genome. DNA is wrapped around protein known as histones and this forms a nucleosome.

  - All somatic cells have 2 of each chromosome in a homologous pair.
  - Homologous chromosomes have the same genes and loci, however the alleles on each chromosome are different

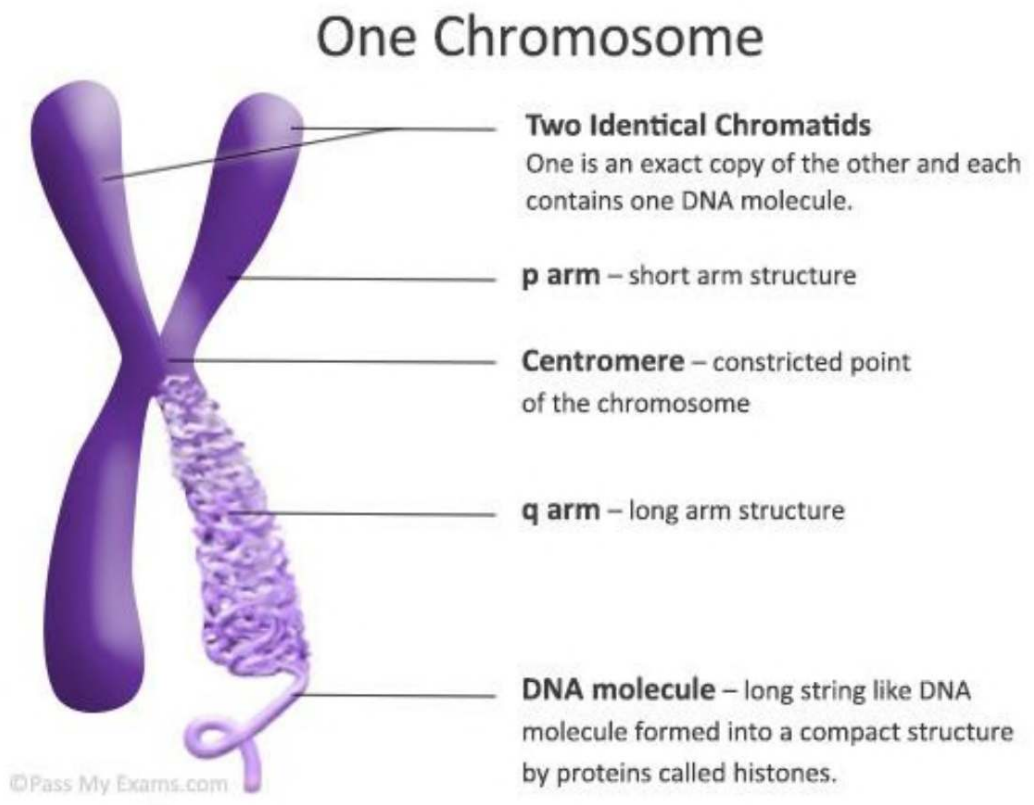
  Chromatin

  Chromatin is the loose, non-condensed version of chromosomes. They remain threadlike, spread apart and loose during the resting state of the cell.

  Chromatin condense to form 1 chromosome or 2 sister chromatids. These are more dense, making them for visible under a light microscope during mitosis.

  Chromatid

  Chromatid is one of the two DNA molecules that make up a chromosome, formed by DNA replication in the S Phase of interphase



Telomere

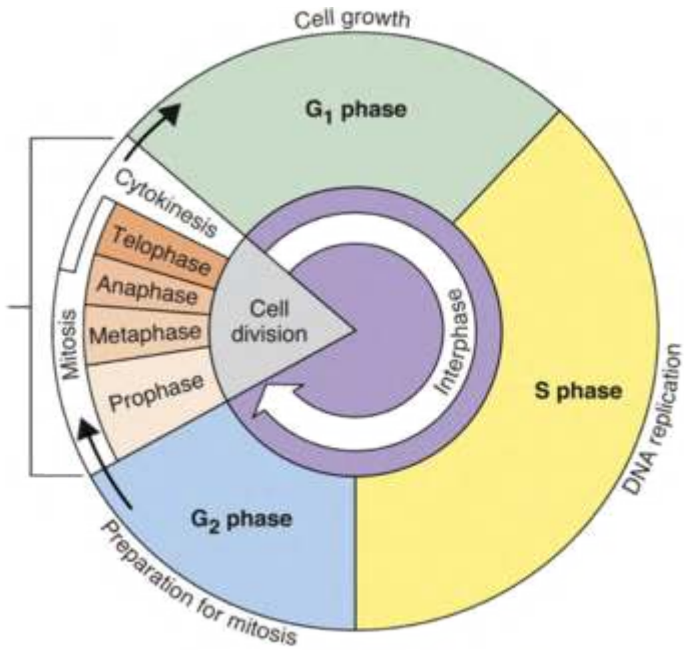
Telomeres - Short, repeated base sequences at the end of the DNA. Ensure preservation of genetic information. Prevents ends from deteriorating/fusing and thus prevents mutations.

Mitosis

▼ What is Mitosis?

Mitosis is the asexual process by which all somatic cells (non-sex cells) divide to produce new cells. This allows organisms to grow and repair and replace damaged cells.

▼ Outline the Cell Cycle



Interphase

- Most of the cell cycle, cells are in interphase where it prepares for mitosis/cell division.

Note: Interphase replicates genetic materal whereas Mitosis replicates cells

- Interphase is broken down into 3 phases - G1 Phase, S Phase and G2 Phase.
    - G1 → Cell Growth
    - S → DNA Replication
    - G2 → Preparation for Mitosis

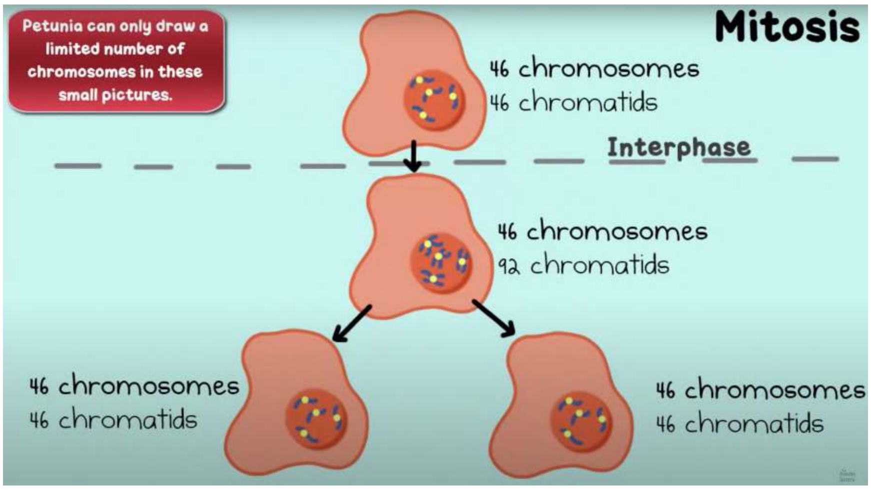
Mitosis

- This is where the cell divides into more cells.
- Broken down into 4 Stages + Cytokinesis - Prophase, Metaphase, Anaphase and Telophase

▼ What happens during the interphase of mitosis?

During interphase, the cell is preparing for division. To do this, it goes through 3 stages.

- G(1) - Increase in cell size, organelles are duplicated, proteins are created (enzymes needed for mitosis), metabolic changes happen.



- S - DNA is duplicated. During replication, 46 chromatids and 46 chromosomes → 92 sister chromatids and 46 chromosomes
- G(2) - Cells ensures everything is ready for mitosis - increases cytoplasmic material, ensures proteins for cytokinesis are present.

G(0) - Mitotic Dormancy. Stays this way usually until an external trigger causes mitosis.

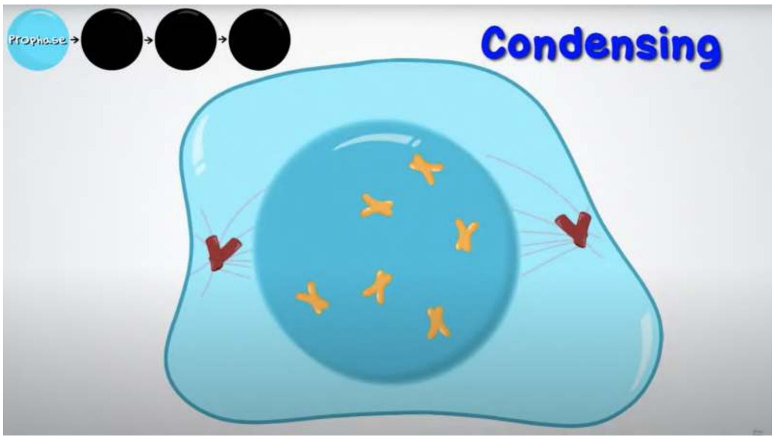
▼ Homologous Chromosomes VS Non-Homologous Chromosomes

For HC, the same genes exist in the same loci but may have different alleles. While for NHC, there are different genes & are usually involved in meiosis.

▼ Describe the use of centrosomes

Centrosomes are centers that contain centrioles that create and release microtubules/spindle fibers in all directions. Each centrosomes have 2 centrioles and these are only present in animal cells. These centrioles **release spindle fibers that connect to the kinetochore of the centromere of the chromosome & pull them apart.**

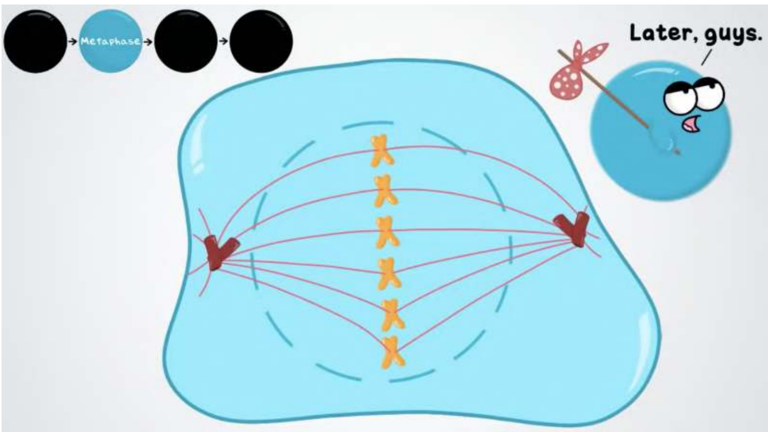
▼ Describe the stages of Mitosis — Prophase



Prophase —

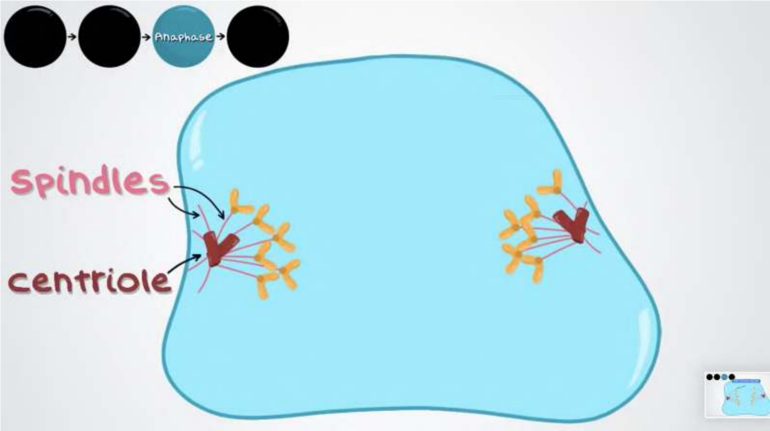
1. Chromatin condense and shorten to form sister chromatids and become visible.
2. Nucleoli disappear.
3. Nuclear membrane/envelope disintegrates.
4. Mitotic Spindle forms and centrosomes move to opposite side of the cell as the mitotic spindle spreads out.
5. Spindle Fibers spread throughout the cell. Polar fibers reach opposite ends as some microtubules attach the kinetochores of the chromosomes.

▼ Describe the stages of Mitosis — Metaphase



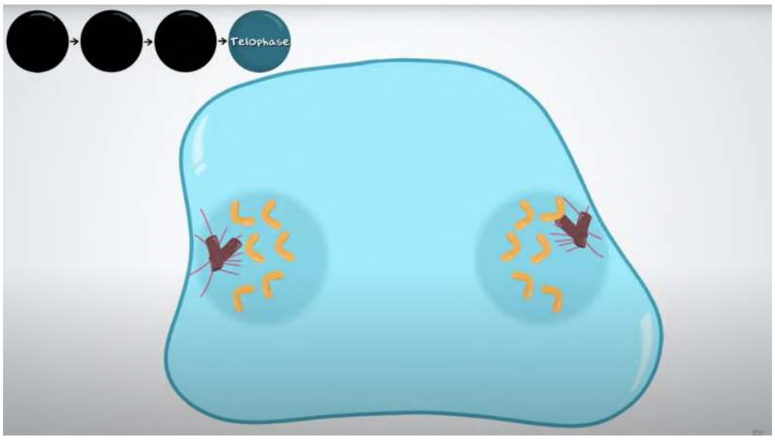
1. The fully developed spindles fibers attach to the kinetochores.
2. Chromosomes align at the equator and centrioles align at opposite ends.

- Describe the stages of Mitosis — Anaphase



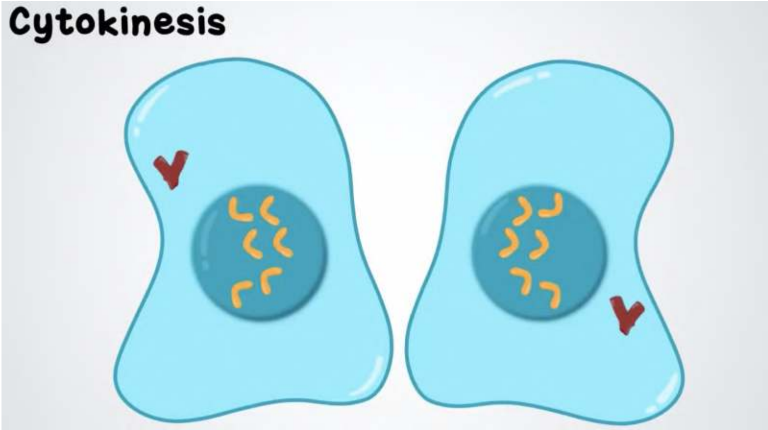
1. Sister chromatids move to opposite sides of the cells, pulled apart by the spindle fibers.
2. Centromere splits
3. Spindle Fiber Contracts
4. The cell elongates as the kinetochores' microtubules lengthen.

- Describe the stages of Mitosis — Telophase



1. Nucleoli begin to form at opposite ends.
2. Nuclear envelope reforms. Nucleoli reformed.
3. Chromosomes uncoil to become chromatin and become enclosed in distinct nucleoli.
4. Spindle disintegrates

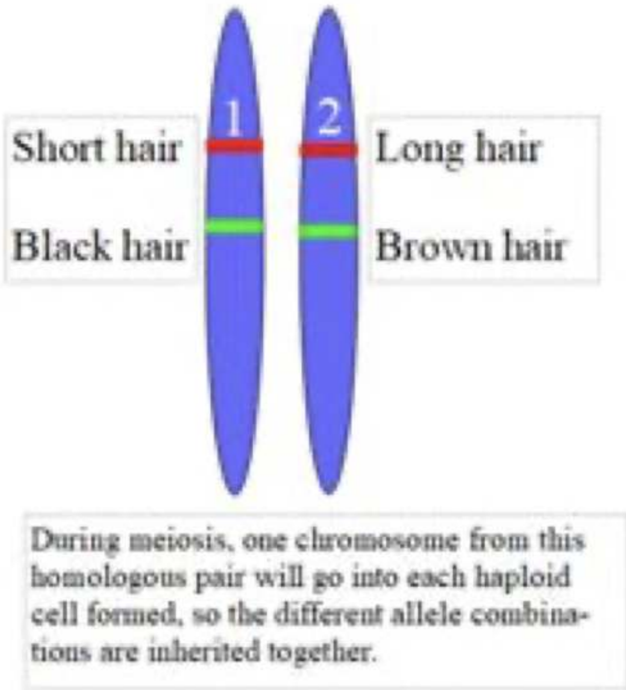
- Explain how cytokinesis occurs in mitosis.



Cytokinesis occurs to finally create 2 new diploid cells in mitosis after telophase. For animal cells, this is done using a cleavage furrow and by pinching with a contractile ring. For plant cells, this is done by the small vesicles aligning in the middle of the cell, forming cell plates & finally creating cell walls in between to create 2 new cells.

Meiosis

▼ Explain Gene Linkage on a Chromosome



If two genes are located on the same chromosome, then they will be **inherited together, i.e. long hair and brown hair would be inherited together or short hair and black hair.** This is because during meiosis, one whole chromosome is passed to the gamete (except in the case of crossing over during meiosis), this is known as **chromosome linkage.**

For instance, if a dog has the genes for hair length and hair colour on the same chromosome; and one of its homologous chromosomes contains the alleles for long and brown hair, and the second chromosome contains the alleles for short and black hair, then the only allele combinations it can pass on to offspring are still long and brown, or short and black - as **one chromosome containing both is passed on to each haploid gamete cell.**

▼ Describe the characteristics of Meiosis.

This only occurs in gametes & produces 4 haploid nuclei and daughter cells from one diploid cell. This process can only occurs in producing gametes.

Meiosis leads to more genetic variation & diversity due to independent assortment & crossing over of chromosomes. Ensures genetic stability by ensuring diploid number of chromosomes from 2 haploid gametes.

(23 HC + 23 HC == 46 DC) (n + n == 2n)

▼ Describe the stages of Meiosis (PMAT x 2)

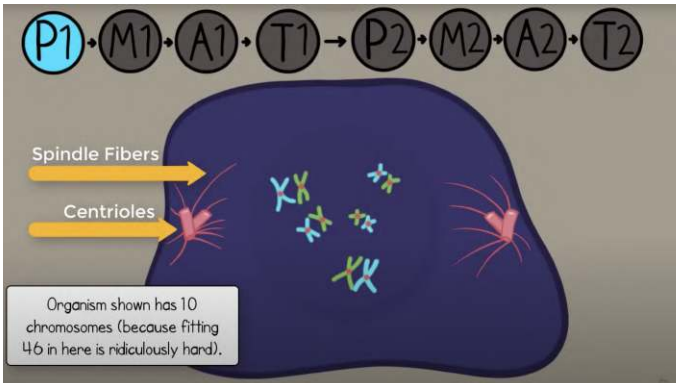
For *interphase*, the stages remain the same.

G1 - Cell Growth, Normal Cell Functions, Production of Organelles.

S - Synthesis of DNA, Chromosomes are duplicated (46 Chromatids → 92 Chromatids, 46 Chromosomes)

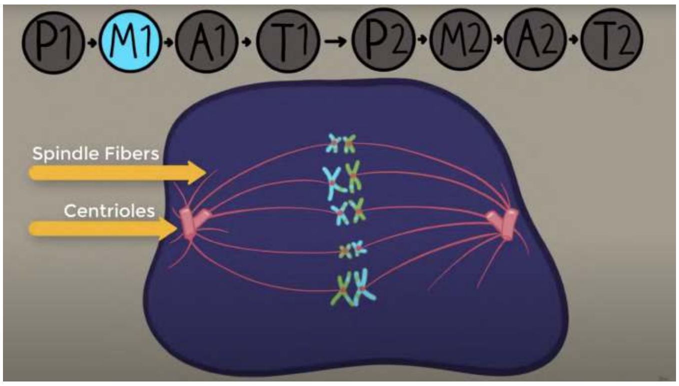
G2 - Ensure everything is ready for mitosis, increases cytoplasmic material, makes sure all the proteins and enzymes for mitosis is present.

For Meiosis 1,



Prophase I —

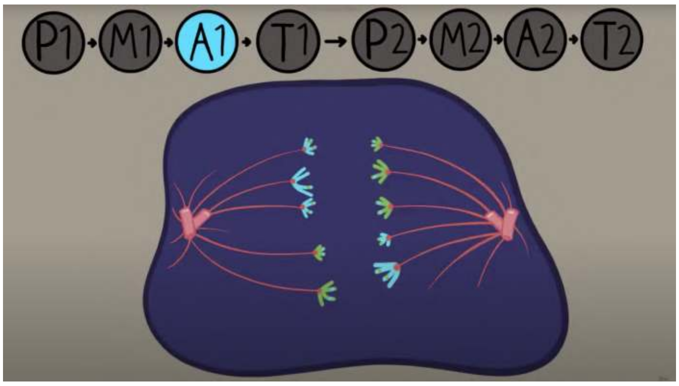
- Chromatin Condenses to form chromosome/sister chromatids.
- Chromosomes line up with their homologous pair (4 Chromatids)
- Centrioles move to the polar ends of the cell and spindle fibers form
- Undergoes Synapsis and Crossing Over - Paternal and Maternal Chromosomes pair up and wrap and wind with eachother tightly to exchange segments of chromosomes (alleles). The region where they exchange segments of DNA is known as the chiasmata.
- Crossing over produces recombinant daughter chromosomes which allow genetic variation.
- Bivalent - Pair of homologous chromosomes close together to form synapses (4 chromatids)



Metaphase I —

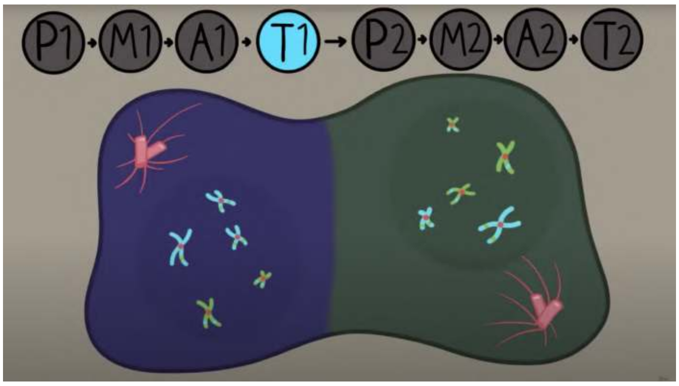
- Allows independent assortment of maternal and paternal chromosomes occur here. Chromosomes can switch orientation so that different combinations can be achieved. 2^pairs = number of sequences possible.

This leads to more genetic variation & stability. Spindle attaches to kinetochores.



Anaphase I —

- Chromosomes are pulled apart by the spindle fibers to both sides of the cell.

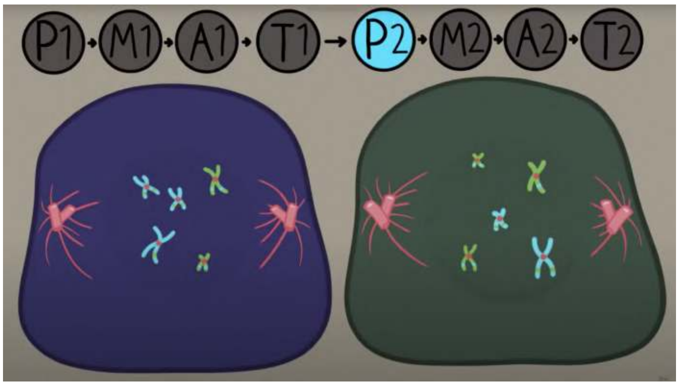


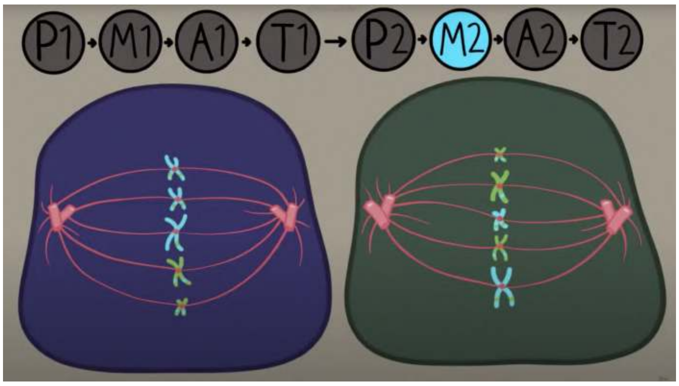
Telophase I —

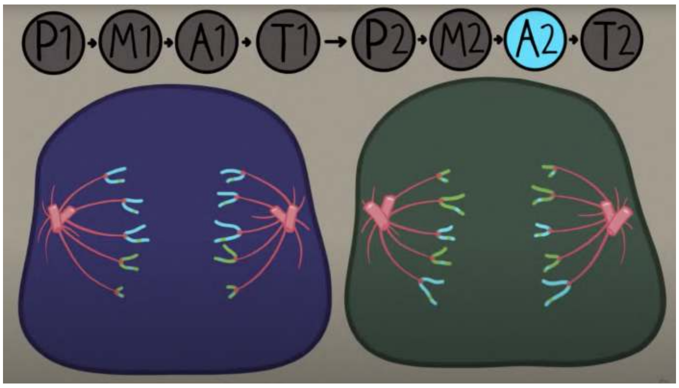
- Nuclei reform around 2 cells.

CYTOKINESIS OCCURS

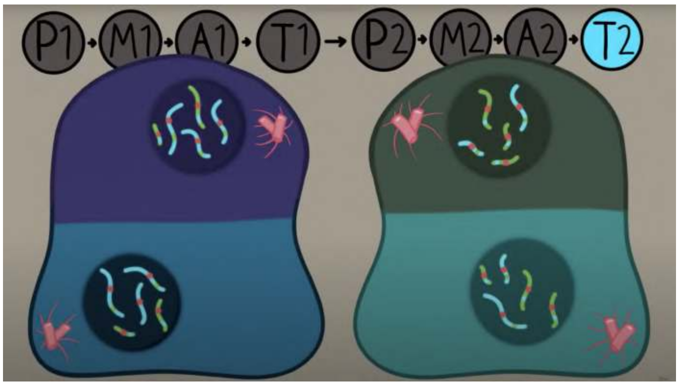
For Meiosis 2,







Prophase II, Metaphase II and Anaphase II occur the same way as in mitosis, but in two diploid cells



Telophase - Nucleoli reform but around 4 haploid cells.

CYTOKINESIS FOR 4 CELLS HAPPEN

Use: formation of gametes.

▼ Explain how Meiosis leads to genetic variation

- In prophase of meiosis I, two pairs of homologous parental chromosomes undergo synapsis/crossing over, where they wind around each other and exchange parts of DNA/chromatids. This produces recombinants/combinations of alleles that gives genetic variation.

- In metaphase of meiosis I, the parental chromosomes undergo independent assortment. This involves the maternal and paternal chromosomes orienting/aliging themselves differently along the equator. This gives different possibilities/combination of alleles in the gamete.

▼ Define the differences between mitosis & meiosis.

S.N.	Differences	Mitosis	Meiosis
1	**Type of Reproduction**	Asexual	Sexual
2	**Genetically**	Similar	Different
3	**Crossing Over**	No, crossing over cannot occur.	Yes, mixing of chromosomes can occur.
4	**Number of Divisions**	One	Two
5	**Pairing of Homologs**	No	Yes
6	**Mother Cells**	Can be either haploid or diploid	Always diploid
7	**Number of Daughter Cells produced**	2 diploid cells	4 haploid cells
8	**Chromosome Number**	Remains the same.	Reduced by half.
9	**Chromosomes Pairing**	Does Not Occur	Takes place during zygotene of prophase I and continue upto metaphase I.
10	**Creates**	Makes everything other than sex cells.	Sex cells only: female egg cells or male sperm cells.
11	**Takes Place in**	Somatic Cells	Germ Cells
12	**Chiasmata**	Absent	Observed during prophase I and metaphase I.
13	**Spindle Fibres**	Disappear completely in telophase.	Do not disappear completely in telophase I.
14	**Nucleoli**	Reappear at telophase	Do not reappear at telophase I.
15	**Steps**	Prophase, Metaphase, Anaphase, Telophase.	(Meiosis 1) Prophase I, Metaphase I, Anaphase I, Telophase I; (Meiosis 2) Prophase II, Metaphase II, Anaphase II and Telophase II.

17	**Cytokinesis**	Occurs in Telophase.	Occurs in Telophase I and in Telophase II.
18	**Centromeres Split**	The centromeres split during anaphase.	The centromeres do not separate during anaphase I, but during anaphase II.
19	**Prophase**	Simple	Complicated
20	**Prophase**	Duration of prophase is short, usually of few hours.	Prophase is comparatively longer and may take days.
21	**Synapsis**	No Synapsis	Synapsis of Homologous chromosomes takes place during prophase.
22	**Exchange of Segments**	Two chromatids of a chromosome do not exchange segments during prophase.	Chromatids of two homologous chromosome exchange segments during crossing over.
23	**Discovered by**	Walther Flemming	Oscar Hertwig
24	**Function**	Cellular reproduction and general growth and repair of the body.	Genetic diversity through sexual reproduction.
25	**Function**	Takes part in healing and repair.	Takes part in the formation of gametes and maintenance of chromosome number.

▼ Define their similarities

Mitosis	Meiosis
Similarities	
Can only occur in eukaryotes	
DNA replication occurs first	
Production of daughter cells based on parent cell's genetic material	
Means of cell replication in plants, animals, and fungi	

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